

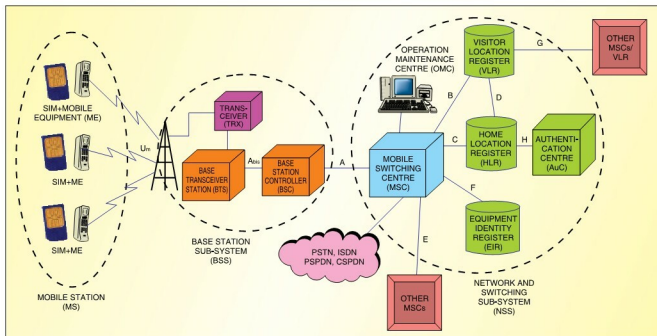


Fig. 1: GSM architecture and network interfaces

Identity (IMSI) used to identify the subscriber to the system, a secret key (Ki) for authentication and algorithms for authentication checks. IMEI and IMSI are independent, thereby allowing personal mobility.

Base station sub-system. The BSS consists of one base station controller (BSC) and one or more base transceiver stations (BTS)—a network component along with a transceiver that serves one cell, provides radio access to the mobile stations and manages the radio access aspects of the system.

The BSC is a network component. It performs radio resource management for the cells under its control. It assigns and releases frequencies and timeslots for all the MS in its area. The BSC performs the inter-cell handover for the MS moving between the BTS in its control. It also reallocates frequencies to the BTS in its area to meet locally heavy demands during peak hours or special events. The BSC also controls the power transmitted by the BTS or MS.

Network and switching sub-system. The NSS combines the call routing switches (MSCs and gateway MSC) with database registers required to

keep track of the subscribers' movements and use of the system. The main role of the MSC is to manage the communications between GSM users and other telecommunication network users (Integrated Services Digital Network (ISDN), Public Switched Telephone Network (PSTN), Packet-Switched Public Data Network (PSPDN) and Circuit-Switched Public Data Network (CSPDN) users).

The MSC performs billing on calls for all the subscribers based in its areas. When a subscriber is on roaming elsewhere, it obtains data for the call-billing from the visited MSC. The MSC specifically performs such functions as paging, resource allocation, location registration and encryption.

Associated registers. The home location register (HLR) contains all the administrative information of each subscriber registered in the corresponding GSM network, along with the current location of the mobile. There is logically one HLR per GSM network, though it may contain several physical HLRs depending on the number of mobile subscribers, capacity of the equipment and the organisation of the network. However, even if the HLR

comprises geographically-separated hardware, it logically forms a single virtual database.

The visitor location register (VLR) is a database serving temporary subscribers within an MSC area. Each MSC in the network has an associated VLR that may serve many MSCs. A mobile station roaming in an MSC area is controlled by the VLR associated with that MSC.

When an MS enters a new location area, it starts a registration procedure. The MSC in charge of that area notices this registration and transfers the identity of the location area where the MS to the VLR is situated. If this MS is not yet registered, the VLR and HLR exchange information to allow proper handling of calls involving the MS.

The authentication centre (AuC) is associated with an HLR and stores an identity key for each mobile subscriber registered with the associated HLR. This key is used to generate data, which is used to authenticate the IMSI. The key is also used to encrypt user-data and cipher communication over the radio path between the mobile station and the network.

The equipment identity register